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Liquid discharge head and liquid discharge apparatus

Abstract

A liquid discharge head includes a discharge port array having a plurality of discharge ports and a supply port array having a plurality of supply ports. The discharge port array and the supply port array extend from a first to a second end of the liquid discharge head. The plurality of discharge ports includes a first-end discharge port closest to the first end and a second-end discharge port closest to the second end. The plurality of supply ports includes a first-end supply port closest to the first end and a second-end supply port closest to the second end. Seen from a position facing openings of the plurality of discharge ports, an end of an opening of the first-end supply port adjacent to the first end is at a position nearer to the first end than an end of an opening of the first-end discharge port adjacent to the first end.

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2017124617	12/2016	JP	N/A

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Background/Summary

BACKGROUND

Field

(1) The present disclosure relates to a liquid discharge head and a liquid discharge apparatus including the liquid discharge head.

Description of the Related Art

(2) The recent ink jet printer field requires liquid discharge apparatuses capable of outputting high-quality print. However, if liquid discharge ports are continuously kept in contact with air for a long time, evaporation of the liquid causes thickening and sticking, which leads to a discharge failure, making it impossible to output high-quality print. To reduce or eliminate the liquid thickening and sticking due to the evaporation of the liquid, known circulating liquid discharge apparatuses circulate liquid.

(3) For example, Japanese Patent Laid-Open No. 2017-124617 discloses a liquid discharge head, as well as a liquid discharge apparatus, including highly densely arranged discharge ports, in which a plurality of common supply channels and a plurality of common collecting channels are provided in the discharge port array to circulate liquid through pressure chambers corresponding to the plurality of discharge ports. FIG. **11** is a schematic diagram of a discharge module **800** of Japanese Patent Laid-Open No. 2017-124617 seen from the liquid discharge direction. FIG. **12** is a schematic plan view of the discharge module **800** exploded into an opening plate **810**, a channel

plate **820**, and a nozzle plate **830**. The opening plate **810** includes supply ports **812** for supplying liquid to the channel plate **820** and collection ports **811** for collecting the liquid from the channel plate **820**. The channel plate **820** includes a common supply channel **822** communicating with the supply ports **812** and a common collecting channel **821** communicating with the collection ports **811**, respectively. The channel plate **820** further includes individual supply channels **824** for supplying the liquid flowing through the common supply channel **822** to discharge ports **831** formed in the nozzle plate **830** and individual collecting channels **823** for collecting the liquid from the discharge ports **831** to the common collecting channel **821**. In other words, by circulating the liquid flowing to the discharge module **800** through the supply ports **812**, the common supply channel **822**, the individual supply channels **824**, the discharge ports **831**, the individual collecting channels **823**, the common collecting channel **821**, and the collection ports **811** in this order, thickening and sticking of the liquid due to evaporation can be reduced or eliminated.
(4) In a conventional liquid discharge apparatus having a liquid discharge head, interruption of circulation of liquid can cause sticking due to evaporation of the liquid, resulting in a discharge failure. When the circulation of the liquid is at a stop, liquid in a certain area may be less likely to be diffused than being evaporated from a discharge port, which can cause liquid sticking at the dead end, resulting in a discharge failure.
(5) In particular, in a conventional serial liquid discharge head that discharges liquid while scanning a medium, if stopped during the scanning because of a trouble during printing, discharge ports may be left in contact with air for a long period of time. Therefore, the liquid sticking and discharge failure described above are more likely to occur.
SUMMARY
(6) The present disclosure provides a liquid discharge head and a liquid discharge apparatus in which liquid sticking and discharge failure can be reduced or eliminated.
(7) According to an aspect of the present disclosure, a liquid discharge head includes a discharge port array in which a plurality of discharge ports configured to discharge liquid is arrayed, a plurality of pressure chambers individually communicating with the plurality of discharge ports, a common supply channel configured to supply liquid to the plurality of pressure chambers, and a supply port array in which a plurality of supply ports configured to supply liquid to the common supply channel is arrayed, wherein the discharge port array and the supply port array extend from a first end of the liquid discharge head to a second end of the liquid discharge head, wherein the plurality of discharge ports in the discharge port array includes a first-end discharge port closest to the first end and a second-end discharge port closest to the second end, wherein the plurality of supply ports in the supply port array includes a first-end supply port closest to the first end and a second-end supply port closest to the second end, and wherein, as seen from a position facing openings of the plurality of discharge ports, an end of an opening of the first-end supply port adjacent to the first end is at a position nearer to the first end than an end of an opening of the first-end discharge port adjacent to the first end.
(8) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram of a liquid discharge apparatus.
(2) FIG. **2**A is a perspective view of the liquid discharge head.
(3) FIG. **2**B is an exploded view of the liquid discharge head.
(4) FIG. **2**C is a bottom view of the liquid discharge head.
(5) FIG. **3**A is a schematic top view of the discharge unit illustrating the circulation paths.

(6) FIG. **3**B is a schematic bottom view of the discharge unit illustrating the circulation paths.
(7) FIG. **4**A is a cross sectional view taken along line IVA-IVA in FIG. **3**A.
(8) FIG. **4**B is a cross sectional view taken along line IVB-IVB in FIG. **3**A.
(9) FIG. **4**C is a cross sectional view taken along line IVC-IVC in FIG. **3**A,
(10) FIG. **5** is a schematic diagram of a discharge module.
(11) FIG. **6** is an exploded schematic diagram of the discharge module.
(12) FIG. **7** is an enlarged view of a first end of the discharge module.
(13) FIG. **8** is an enlarged view of a second end of the discharge module.
(14) FIG. **9** is a schematic diagram of the discharge module.
(15) FIG. **10** is a schematic diagram of a discharge module of a second embodiment.
(16) FIG. **11** is a schematic diagram of a discharge module in related art.
(17) FIG. **12** is an exploded schematic diagram of the discharge module in the related art.

DESCRIPTION OF THE EMBODIMENTS

(18) Embodiments of the present disclosure will be described hereinbelow with reference to the drawings. It is to be understood that the following embodiments do not limit the present disclosure and that not all of combinations of the features described in the embodiments are absolutely necessary for the solution of the present disclosure. Like components are denoted by like reference signs.

First Embodiment

(19) FIG. **1** is a schematic perspective view of a liquid discharge apparatus **50** including a liquid discharge head **1**. The liquid discharge apparatus **50** of this embodiment is a serial ink-jet printing apparatus that discharges ink, or liquid, while moving the liquid discharge head **1** to print on a recording medium P. Another example is a what-is-called full-line liquid discharge head including discharge ports across the width of the recording medium P so as to be capable of discharge across the width of the recording medium P without moving in a main scanning direction, described below.

(20) The liquid discharge head **1** is mounted on a mount (a carriage **60**). The carriage moves back and forth in the main scanning direction (X-direction) along a guide shaft **51**. The recording medium P is conveyed in a sub-scanning direction (Y-direction) intersecting (in this example, at right angles) the main scanning direction by conveying rollers **55**, **56**, **57**, and **58**. The liquid discharge head **1** is detachably attached to the carriage **60** by the user.

(21) The liquid discharge head **1** includes a circulation unit **54** and a discharge unit **3**, described below. The discharge unit **3** includes a plurality of discharge ports for discharging liquid and energy generating elements that generate discharge energy for discharging liquid from the individual discharge ports, the specific configuration of which will be described below.

(22) The liquid discharge apparatus **50** includes an ink tank **2**, which is an ink supply source, and an external pump **21**. The ink stored in the ink tank **2** is supplied to the circulation unit **54** through ink supply tubes **59** with the driving force of the external pump **21**.

(23) The liquid discharge apparatus **50** forms a predetermined image on the recording medium P by repeating a recording scanning operation in which the liquid discharge head **1** mounted on the carriage **60** discharges ink while moving back and forth in the main scanning direction and a conveying operation of conveying the recording medium P in the sub-scanning direction. The liquid discharge head **1** of this embodiment is capable of discharging four kinds of ink, black (K), cyan (C), magenta (M), and yellow (Y), and can record a full-color image with the inks. The ink that the liquid discharge head **1** can discharge is not limited to the above four kinds of ink. The present disclosure is also applicable to liquid discharge heads for discharging other kinds of ink. In other words, the kind and number of inks discharged from the liquid discharge head are not limited. In the case where a cap member (not shown) is disposed off the conveying path of the recording medium P, when no printing operation is to be performed, the cap member is relatively moved to a position where the face of the liquid discharge head **1** is to be covered to prevent the discharge

ports from drying, or a sucking operation is performed for charging or recovery.
(24) FIGS. **2**A to **2**C are schematic diagrams of the liquid discharge head **1**. FIG. **2**A is a schematic perspective view of the liquid discharge head **1**. FIG. **2**C is a diagram of the liquid discharge head **1** seen from arrow IIC in FIG. **2**A. The outer shape of the liquid discharge head **1** is formed by a casing **110** and a supporting member **210**. FIG. **2**B is an exploded perspective view of the liquid discharge head **1**. The liquid discharge head **1** includes four circulation units **54** corresponding to four kinds of ink. The circulation unit **54** includes circulation units **54***a* to **54***d* corresponding to the individual inks. The circulation units **54***a* to **54***d* are connected to the casing **110**. The casing **110** has a joint surface **111** for receiving ink from the printing apparatus. The joint surface **111** is provided with joints **111***a* to **111***d* communicating with the circulation units **54***a* to **54***d*, respectively. In mounting the liquid discharge head **1** to the printing apparatus, the ink supply tubes **59** corresponding to the individual inks are connected from the printing apparatus to the joints **111***a* to **111***d*. The inks supplied through the ink supply tubes **59** pass through the joints **111***a* to **111***d* of the casing **110** to the circulation units **54***a* to **54***d*. A plurality of circulation units **54** may be provided for the same kind of liquid.
(25) In FIG. **2**B, the liquid discharge head **1** includes the circulation unit **54**. The circulation unit **54** includes a pressure adjusting unit (not shown) and a pump. An example of the pump is a diaphragm pump. Any pump capable of circulating liquid may be employed. The liquid discharge head **1** does not have to include the circulation unit **54**. In other words, a circulation unit provided outside the liquid discharge head **1** may be used to circulate the liquid in the liquid discharge head **1** to the external circulation unit.
(26) The bottom **113** of the casing **110** connects to a supporting member **210** on which a discharge module **300**, to be described later, is mounted. The ink supplied to the circulation units **54***a* to **54***d* is supplied to the supporting member **210** through the casing **110**. In FIG. **2**C, two discharge modules **300** are joined to the supporting member **210**. Alternatively, one or three or more discharge modules **300** may be joined. The discharge modules **300** and the supporting member **210** are bonded with an adhesive. The discharge modules **300** includes a silicon substrate with a thickness of 0.5 mm to 1 mm and a plurality of energy generating elements **36** (FIG. **4**A) provided on one surface of the silicon substrate and for discharging liquid. In this embodiment, heating resistive elements are used as the energy generating elements **36**, which generate heat using electric power supplied from a printed circuit board **230** via an electric wiring board **220**. The silicon substrate includes a plurality of pressure chambers corresponding to the heating resistive elements and a plurality of discharge ports for discharging ink formed using a photolithographic technique. In this embodiment, the heating resistive elements are used as the energy generating elements **36**. This is given for mere illustrative purposes. In other words, a liquid discharge method of using piezoelectric elements as energy generating elements or another discharge method may be employed.
(27) FIGS. **3**A and **3**B are schematic diagrams illustrating the circulation path in the discharge unit **3**. FIG. **3**A is an exploded perspective view of the discharge unit **3** seen from the supporting member **210**. FIG. **3**B is an exploded perspective view of the discharge unit **3** seen from the nozzle plate **330**. The ink channel in the discharge unit **3** is constituted by the supporting member **210** and the discharge module **300**. The arrows in FIGS. **3**A and **3**B indicate the circulation flow of the liquid in the discharge unit **3**. The solid arrows indicate the flow of liquid supplied from the supporting member **210** to the discharge module **300**. The dashed arrows indicate the flow of liquid collected from the discharge module **300** to the supporting member **210**. The ink flow is described for one color only, but the same applies to the other colors.
(28) The supporting member **210** includes a plurality of supporting member supply ports **212** and a plurality of supporting member collection ports **211**. The liquid supplied from the liquid discharge apparatus **50** to the liquid discharge head **1** or the liquid supplied from the circulation unit **54** is supplied to the discharge module **300** through the supporting member supply ports **212**. The liquid

not discharged by the discharge module **300** is collected to the circulation unit **54** through the supporting member collection ports **211**. If the liquid discharge head **1** does not include the circulation unit **54**, the liquid may be collected by a circulation unit outside the liquid discharge head **1**.

(29) The discharge module **300** is a unit serving as a liquid discharge head, and the discharge module **300** itself is also referred to as a liquid discharge head. The discharge module **300** is a lamination of an opening plate **310**, a channel plate **320**, and a nozzle plate **330** layered in this order from the supporting member **210**. These plates are superposed and joined so that the liquid channels communicate to constitute the discharge module **300** and are supported by the supporting member **210**.

(30) The opening plate **310** includes, on a surface adjacent to the supporting member **210**, a supply port array in which supply ports **312** communicating with the supporting member supply ports **212** are arrayed and a collection port array in which collection ports **311** communicating with the supporting member collection ports **211** are arrayed. The supply port array is an array of a plurality of supply ports for supplying liquid to one common supply channel **322**, described later. Likewise, the collection port array is an array of a plurality of collection ports for collecting liquid from one common collecting channel **321**, described later.

(31) Although FIGS. **3**A and **3**B illustrate multiple supply ports **312** and collection ports **311**, the number of the supply port **312** and the collection ports **311** may be one, or one of them may be multiple, and the other may be only one.

(32) The channel plate **320** includes, on a surface adjacent to the opening plate **310**, a common supply channel **322** communicating with the supporting member supply ports **212** and a common collecting channel **321** communicating with the supporting member collection ports **211**. The channel plate **320** includes, on a surface adjacent to the nozzle plate **330**, individual supply channels **324** each connecting the common supply channel **322** with a pressure chamber **325** (FIGS. **4**A to **4**C) formed in the nozzle plate **330** and individual collecting channels **323** each connecting the common collecting channel **321** with the pressure chamber **325**. The nozzle plate **330** has a discharge port array in which a plurality of discharge ports **331** that discharges liquid is arrayed. The plurality of discharge ports **331** communicates with the plurality of pressure chambers **325**, respectively. The pressure generated by the energy generating element **36** acts on the liquid in each pressure chamber **325** to cause the liquid to be discharged from the discharge port **331**. The liquid not discharged is collected from the pressure chamber **325** into the common collecting channel **321** and the collection port **311** through the individual collecting channel **323**.

(33) In other words, the liquid in the discharge module **300** circulates through the supply port **312**, the common supply channel **322**, the individual supply channel **324**, the pressure chamber **325**, the individual collecting channel **323**, the common collecting channel **321**, and the collection port **311** in this order. The presence of the common supply channel **322** and the individual supply channel **324** allows the liquid flowing from the supply port **312** to be directed to the position where the discharge port **331** is formed. Likewise, the presence of the common collecting channel **321** and the individual collecting channel **323** allows the liquid not discharged from the discharge port **331** to be directed to the position where the collection port **311** is formed. The supply ports **312** and the collection ports **311** of the opening plate **310** are formed in a staggered arrangement. If not in the staggered arrangement, the supply ports **312** and the collection ports **311** may be arranged in parallel in a straight line, as shown in FIG. **10**, described later. This increases the length of the liquid discharge head in the X-direction (lateral direction). For this reason, the supply ports **312** and the collection ports **311** may be formed in a staggered arrangement in the viewpoint of production cost.

(34) As shown in FIG. **10**, described later, the supply ports **312** and the collection ports **311** may be formed in parallel.

(35) FIGS. **4**A to **4**C are cross-sectional views of the discharge unit **3** illustrating ink flows at

different portions. FIG. **4**A illustrates a cross section taken along line IVA-IVA in FIG. **3**A, where the supporting member supply ports **212** and the supply ports **312** of the discharge unit **3** communicate with each other. FIG. **4**B illustrates a cross section taken along line IVB-IVB in FIG. **3**A, where the supporting member collection ports **211** and the collection ports **311** of the discharge unit **3** communicate with each other. FIG. **4**C illustrates a cross section taken along line IVC-IVC in FIG. **3**A, where the collection ports **311** and the supply ports **312** do not communicate with the channels of the supporting member **210**.

(36) The supply channel for supplying liquid supplies the liquid from a portion where the supporting member supply port **212** of the supporting member **210** and the supply port **312** of the opening plate **310** are superposed and communicate with each other, as shown in FIG. **4**A. The collection channel for collecting liquid collects the liquid from a portion where the supporting member collection port **211** of the supporting member **210** and the collection port **311** of the opening plate **310** are superposed and communicates with each other, as shown in FIG. **4**B. When the circulation of the liquid is at a stop, the liquid supplied from the collection channel to the pressure chamber **325** is more likely to be diffused than being evaporated from the discharge port **331**. As shown in FIG. **4**C, the discharge unit **3** partially includes an area of the opening plate **310** having no opening. In such an area, no liquid is supplied and collected between the supporting member **210** and the discharge module **300**. Liquid is supplied in the area in which the supply ports **312** are provided, as shown in FIG. **4**A, and is collected in the area where the collection ports **311** are provided, as shown in FIG. **4**B.

(37) Since the discharge ports **331** are disposed in the paths connecting the individual supply channels **324** and the individual collecting channels **323**, flows from the individual supply channels **324** to the individual collecting channels **323** are generated in the pressure chambers **325** in the vicinity of the discharge ports **331**. The flows are generated over the length from the first end to the second end of the discharge port array. This increases the liquid circulation efficiency to keep the liquid in the pressure chambers **325** constantly fresh, thereby preventing issues, such as thickening and sticking due to evaporation.

(38) The known opening plate **810** shown in FIG. **11** has the supply ports **812** and the collection ports **811** formed in a staggered arrangement, as described above. This produces a dead end (area B) in an intermediate point of the channel from the supply ports **812** to the discharge ports **831** of the common supply channel **822**.

(39) FIG. **5** is a schematic diagram of the discharge module **300** of this embodiment seen from the supporting member **210** (opening plate **310**). FIG. **6** is a schematic plan view of the discharge module **300** exploded into the opening plate **310**, the channel plate **320**, and the nozzle plate **330**. FIG. **7** is a schematic diagram of a first end of the discharge module **300** shown in FIG. **5** in an enlarged view. Also in this embodiment, the plurality of supply ports **312** and the plurality of collection ports **311** are formed in a staggered arrangement in the viewpoint of production cost.

(40) In the present disclosure, the discharge port array of the discharge ports **331** and the supply port array of the plurality of supply ports **312** extend in the Y-direction from a first end **300***a* to a second end **300***b* of the discharge module **300** (liquid discharge head). The plurality of discharge ports **331** in the discharge port array includes a first-end discharge port **331***c* disposed closest to the first end **300***a* and a second-end discharge port **331***d* disposed closest to the second end **300***b*. The plurality of supply ports **312** in the supply port array includes a first-end supply port **312***c* disposed closest to the first end **300***a* and a second-end supply port **312***d* disposed closest to the second end **300***b*. In this embodiment, a partial area of the opening of the first-end supply port **312***c* is formed outside an end **331***a* of the opening of the first-end discharge port **331***c* in the Y-direction. In other words, an end **312***a* of the opening of the first-end supply port **312***c* adjacent to the first end **300***a* (hereinafter referred to as “first-side end”) is nearer to the first end **300***a* than a first-side end **331***a* of the opening of the first-end discharge port **331***c* seen from a position facing the opening of the discharge ports **331**. The end of the opening of the supply port and the end of the opening of the

discharge port are respectively closest to the first end **300***a* or the second end **300***b* of the discharge module **300** among the respective openings. This is the same for the collection ports **311** as for the supply ports **312**.

(41) In FIG. **7**, the supply ports **312** and the collection ports **311** are rectangular. However, this is given for illustrative purposes, and any other shape that functions as the supply ports **312** and the collection ports **311** may be employed. Likewise, the shape of the discharge ports **331** need not be circular and may be any shape that allows liquid to be discharged.

(42) The above configuration has no dead end (the area B in FIG. **11**) at an intermediate point of the channel from the supply ports **312** to the discharge ports **331** in the common supply channel **322**. This allows the supply channel to have only the area A between the two supply ports **312**, where liquid is supplied from the two supply ports **312**. In the case where the collection channel is disposed as in FIG. **5**, only the area C between the two collection ports **311**, where the liquid is collected to the two collection ports **311**, is formed. As a result, a dead end where the liquid circulation flow tends to stagnate is not formed. A dead end where diffusion of liquid is less likely to occur than evaporation from the discharge ports **331** is not present in both the supply channel and the collection channel. This configuration prevents a liquid flow from being blocked even when circulation of liquid circulation is at a stop. This configuration reduces or eliminates liquid sticking and discharge failure.

(43) Here, the common supply channel **322** communicating with the supply ports **312** may extend outward in the Y-direction from the first end or the second end of the discharge port array. In other words, the length of the common supply channel **322** may be larger than the length T of the discharge port array in the direction in which the supply port array extends.

(44) With this configuration of the supply ports **312** and the common supply channel **322**, the supply ports **312** and the common supply channel **322** located outside the discharge port array are present also at a position not overlapping with the energy generating elements **36** that generate heat to discharge liquid in the vertical direction. This allows the liquid not heated by the energy generating elements **36** to be supplied to the pressure chambers **325** through the supply ports **312** and the common supply channel **322** located outside the discharge port array. Accordingly, fresh liquid at near room temperature not heated by the energy generating elements **36** can be supplied to the ends of the discharge ports **331**.

(45) Even without the dead end (area B) in FIG. **11**, the liquid flow tends to stagnate at the ends of the common supply channel **322**. Here, the ends of the common supply channel **322** are channel walls of the common supply channel **322** closest to the opposite ends of the discharge module **300**. The ends of the common supply channel **322** extend outward in the Y-direction from the discharge port array, so that the discharge ports **331** and the ends of the common supply channel **322** are not aligned in the vertical direction. This reduces the effect of the liquid sticking at the ends of the common supply channel **322** on the discharge of liquid from the discharge ports **331**.

(46) In FIG. **7**, the entire area of the first-end supply port **312***c* is formed outside the end **331***a* of the opening of the first-end discharge port **331***c* in the Y-direction. In other words, an end **312***b* of the opening of the first-end supply port **312***c* adjacent to the second end **300***b* (hereinafter referred to as “second-side end”) is formed at a position nearer to the first end **300***a* than the first-side end **331***a* of the opening of the first-end discharge ports **331***c*. This configuration also does not have a dead end where the liquid flow tends to stagnate, preventing liquid sticking and discharge failure. This configuration also increases the amount of liquid, of the liquid to be supplied to the discharge ports **331**, not heated by the heat generated by the energy generating elements **36**.

(47) FIG. **8** is an enlarged schematic diagram of the second end **300***b* of the discharge module **300** in FIG. **5**.

(48) The second end **300***b*, with the same configuration as the first end **300***a*, increases the effect of the present disclosure. In other words, a partial area of the opening of the second-end supply port **312***d* may be formed outside an end **331***b* of the opening of the second-end discharge port **331***d* in

the Y-direction. In other words, the second-side end **312***b* of the opening of the second-end supply port **312***d* may be formed at a position nearer to the second end **300***b* than the second-side end **331***b* of the opening of the second-end discharge port **331***d*.
(49) The entire area of the opening of the second-end supply port **312***d* may be formed the end **331***b* of the opening of the second-end discharge port **331***d* in the Y-direction. In other words, the first-side end **312***a* of the opening of the second-end supply port **312***d* is formed at a position nearer to the second end **300***b* than the second-side end **331***b* of the opening of the second-end discharge port **331***d*.
(50) In this embodiment, the collection ports **311**, with the same configuration as the supply ports **312**, increase the effect of the present disclosure. In other words, the collection ports **311**, formed outside the discharge port array, also have no dead end in the common collecting channel **322** between the collection ports **311** and the discharge ports **331**. This allows the liquid in the area C to be collected to the two collection ports **311** during liquid circulation. When the circulation is at a stop, only the area C where the liquid is supplied from the two collection ports **311** because of the diffusion of the liquid is formed from the collection ports **311** to the discharge ports **331** in the common collecting channel **322**. This reduces the possibility of blocking of the liquid flow, reducing or eliminating liquid sticking and discharge failure.
(51) The plurality of collection ports **311** in the collection port array includes a first-end collection port **311***c* closest to the first end **300***a* and a second-end collection port **311***d*. A partial area of the opening of the first-end collection port **311***c* may be formed outside the end **331***a* of the opening of the first-end discharge port **331***c* in the Y-direction. In other words, the first-side end **311***a* of the opening of the first-end collection port **311***c* may be formed nearer to the first end **300***a* than the first-side end **331***a* of the opening of the first-end discharge port **331***c*.
(52) Furthermore, the entire area of the opening of the first-end collection port **311***c* may be formed outside the end **331***a* of the opening of the first-end discharge port **331***c* in the Y-direction. In other words, a second-side end **311***b* of the opening of the first-end collection port **311***c* may be at a position nearer to the first end **300***a* than the first-side end **331***a* of the opening of the first-end discharge port **331***c*.
(53) The collection ports **311** on the second end **300***b*, with the same configuration as on the first end **300***a*, further improve the effect of the present disclosure. In other words, a partial area of the opening of the second-end collection port **311***d* may be formed outside the end **331***b* of the opening of the second-end discharge port **331***d* in the Y-direction. In other words, the second-side end **311***b* of the opening of the second-end collection port **311***d* may be formed at a position nearer to the second end **300***b* than the second-side end **331***b* of the opening of the second-end discharge port **331***d*.
(54) Furthermore, the entire area of the opening of the second-end collection port **311***d* may be formed outside the end **331***b* of the opening of the second-end discharge port **331***d* in the Y-direction. In other words, a first-side end **311***a* of the opening of the second-end collection port **311***d* is formed at a position nearer to the second end **300***b* than the second-side end **331***b* of the opening of the second-end discharge port **331***d*.
(55) Even without the dead end (area B) in FIG. **11**, the liquid flow tends to stagnate at the ends of the common collecting channel **321**, as in the common supply channel **322**. Here, the ends of the common collecting channel **321** are channel walls of the common collecting channel **321** closest to the opposite ends of the discharge module **300**. In FIG. **9**, the ends of the common collecting channel **321** extend outward in the Y-direction from the discharge port array. In other words, the length of the common collecting channel **321** is larger than the length T of the discharge port array in the direction in which the supply port array extends, so that the discharge ports **331** and the ends of the common collecting channel **321** are not aligned in the vertical direction. This reduces the possibility of blocking of the liquid flowing in the common collecting channel **321**.
(56) The configuration in which the collection ports **311** at the opposite ends of the collection port

array are nearer to the opposite ends of the discharge module **300** than the discharge ports **331** produces the effect of easily removing bubbles and foreign particles generated at the ends of the discharge port array, the common collecting channel, and the collection port array.
(57) In FIG. **5**, the openings of the plurality of supply ports and the openings of the plurality of collection ports are not aligned in the direction in which the discharge port array extends. The number of supply ports for supplying liquid to one common supply channel **322** is six, end the number of collection ports for collecting the liquid from one common collecting channel **321** is five. However, the numbers of the supply ports **312** and the collection ports **311** are not limited to the ones described above. In other words, the number of supply ports **312** and the number of collection ports **311** may be any number.
(58) Since the liquid that has passed through the supply ports **312** are generally discharged from the discharge ports **331**, the liquid flowing through the collection ports **311** is less than the liquid flowing through the supply ports **312**. For this reason, the number of supply ports **312** may be larger than the number of collection ports **311**. In this case, the common supply channel **322** may be longer than the common collecting channel **321** in the direction in which the discharge port array extends to prevent a dead end from being formed in the common collecting channel **321**.
(59) The number of supply ports **312** communicating with one common supply channel **322** and the number of collection ports **311** communicating with one common collecting channel **321** may be the same for symmetry of the design. The supply ports **312** and the collection ports **311** may be formed in a staggered arrangement at positions where the opposite ends of the common supply channel **322** and the opposite ends of the common collecting channel **321** are not aligned in the direction in which the discharge port array extends.
(60) In this embodiment, the supply ports **312** and the collection ports **311**, the common supply channel **322** and the common collecting channel **321**, the individual supply channels **324** and the individual collecting channels **323** may be formed at inverted positions. Also in this case, forming at least one of the supply ports **312** and the collection ports **311** nearer to the opposite ends of the discharge module **300** than the discharge port array prevents a dead end to be formed, providing the effect of the present disclosure.
(61) In a liquid discharge apparatus including a serial liquid discharge head, if a trouble such as a paper jam occurs during printing to stop the liquid discharge head halfway through scanning, the power source is automatically turned off for safety to stop the circulation of the liquid. If the issue cannot be immediately addressed, the off state is left, and the discharge ports of the liquid discharge head are left in contact with air, which may cause liquid thickening and sticking due to evaporation, resulting in a discharge failure. Accordingly, the liquid discharge head of an embodiment of the present disclosure may be incorporated in a liquid discharge apparatus including a serial liquid discharge head.
(62) The above configuration prevents a dead end from being formed in a common channel from the supply port or the collection port to the discharge port, thereby reducing or eliminating sticking of liquid and a discharge failure.
Second Embodiment
(63) The configuration of a discharge module **300** according to a second embodiment of the present disclosure will be described. In the following description, only the difference from the first embodiment will be mainly described, and description of the same parts as in the first embodiment will be omitted.
(64) FIG. **10** is a schematic diagram of the discharge module **300** of the second embodiment. In the second embodiment, the supply ports **312** and the collection ports **311** are arranged in parallel. In other words, the openings of the plurality of supply ports and the openings of the plurality of collection ports are at least partly aligned in the direction in which the discharge port array extends. Also with this configuration, forming at least one of the supply ports **312** and the collection ports **311** nearer to the opposite ends of the discharge module **300** than the discharge port array prevents

a dead end to be formed, providing the effect of the present disclosure. This configuration allows the number of supply ports **312** or collection ports **311** communicating with one common supply channel **322** or one common collecting channel **321** to be increased, making it easier for the liquid to flow, thereby reducing liquid sticking and discharge failure.
(65) With the above configuration, even if the openings of the plurality of supply ports and the openings of the plurality of collection ports are aligned in the direction in which the discharge port array extends, no dead end is formed, thereby reducing liquid sticking and discharge failure.
(66) Combinations of the configurations of the above embodiments are also applicable.
(67) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(68) This application claims the benefit of Japanese Patent Application No. 2022-091391, filed Jun. 6, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A liquid discharge head comprising: a discharge port array in which a plurality of discharge ports configured to discharge liquid is arrayed; a plurality of pressure chambers respectively communicating with the plurality of discharge ports; a common supply channel configured to supply liquid to the plurality of pressure chambers; and a supply port array in which a plurality of supply ports configured to supply liquid to the common supply channel is arrayed, wherein the discharge port array and the supply port array extend from a first end of the liquid discharge head to a second end of the liquid discharge head, wherein the plurality of discharge ports in the discharge port array includes a first-end discharge port closest to the first end and a second-end discharge port closest to the second end, wherein the plurality of supply ports in the supply port array includes a first-end supply port closest to the first end and a second-end supply port closest to the second end, wherein, as seen from a position facing openings of the plurality of discharge ports, an end of an opening of the first-end supply port adjacent to the first end is at a position nearer to the first end than an end of an opening of the first-end discharge port adjacent to the first end, and wherein at least one of: an end of an opening of the first-end supply port adjacent to the second end is at a position nearer to the first end than an end of the opening of the first-end discharge port adjacent to the first end, and an end of an opening of the second-end supply port adjacent to the first end is at a position nearer to the second end than an end of an opening of the second-end discharge port adjacent to the second end.
2. The liquid discharge head according to claim 1, wherein an end of an opening of the second-end supply port adjacent to the second end is at a position nearer to the second end than the end of an opening of the second-end discharge port adjacent to the second end.
3. The liquid discharge head according to claim 1, further comprising: a common collecting channel configured to collect liquid from the plurality of pressure chambers; and a collection port array in which a plurality of collection ports configured to collect liquid from the common collecting channel is arrayed, wherein the plurality of collection ports in the collection port array includes a first-end collection port at the first end and a second-end collection port at the second end.
4. The liquid discharge head according to claim 3, wherein an end of an opening of the first-end collection port adjacent to the first end is at a position nearer to the first end than the end of the opening of the first-end discharge port adjacent to the first end.
5. The liquid discharge head according to claim 4, wherein an end of an opening of the first-end collection port adjacent to the second end is at a position nearer to the first end than the end of the opening of the first-end discharge port adjacent to the first end.

6. The liquid discharge head according to claim 5, wherein an end of an opening of the second-end collection port adjacent to the second end is at a position nearer to the second end than an end of an opening of the second-end discharge port adjacent to the second end.
7. The liquid discharge head according to claim 6, wherein an end of an opening of the second-end collection port adjacent to the first end is at a position nearer to the second end than the end of the opening of the second-end discharge port adjacent to the second end.
8. The liquid discharge head according to claim 3, wherein openings of the plurality of supply ports and openings of the plurality of collection ports are not aligned in a direction in which the discharge port array extends.
9. The liquid discharge head according to claim 8, wherein the common supply channel is longer than the common collecting channel in the direction in which the discharge port array extends.
10. The liquid discharge head according to claim 8, wherein the common supply channel is longer than the discharge port array in the direction in which the supply port array extends.
11. The liquid discharge head according to claim 10, wherein the common collecting channel is longer than the discharge port array in the direction in which the supply port array extends.
12. The liquid discharge head according to claim 3, wherein opposite ends of the common supply channel are not aligned with opposite ends of the common collecting channel in a direction in which the discharge port array extends.
13. The liquid discharge head according to claim 3, wherein at least part of openings of the plurality of supply ports and openings of the plurality of collection ports are aligned in a direction in which the discharge port array extends.
14. A liquid discharge apparatus comprising: a liquid discharge head; and a mount on which the liquid discharge head is mounted, wherein the mount is configured to move back and forth with respect to a recording medium, wherein the liquid discharge head includes: a discharge port array in which a plurality of discharge ports configured to discharge liquid is arrayed, a plurality of pressure chambers individually communicating with the plurality of discharge ports, a common supply channel configured to supply liquid to the plurality of pressure chambers, a supply port array in which a plurality of supply ports configured to supply liquid to the common supply channel is arrayed; a common collecting channel configured to collect liquid from the plurality of pressure chambers, a collection port array in which a plurality of collection ports configured to collect liquid from the common collecting channel is arrayed, and a pump configured to circulate liquid, wherein the discharge port array and the supply port array extend from a first end of the liquid discharge head to a second end of the liquid discharge head, wherein the plurality of discharge ports in the discharge port array includes a first-end discharge port closest to the first end and a second-end discharge port closest to the second end, wherein the plurality of supply ports in the supply port array includes a first-end supply port closest to the first end and a second-end supply port closest to the second end, wherein, as seen from a position facing openings of the plurality of discharge ports, an end of an opening of the first-end supply port adjacent to the first end is at a position nearer to the first end than an end of an opening of the first-end discharge port adjacent to the first end, wherein the plurality of collection ports in the collection port array includes a first-end collection port at the first end and a second-end collection port at the second end, and wherein an end of an opening of the second-end supply port adjacent to the first end is at a position nearer to the second end than the end of the opening of the second-end discharge port adjacent to the second end.
15. A liquid discharge head comprising: a discharge port array in which a plurality of discharge ports configured to discharge liquid is arrayed; a plurality of pressure chambers individually communicating with the plurality of discharge ports; a common supply channel configured to supply liquid to the plurality of pressure chambers; a supply port array in which a plurality of supply ports configured to supply liquid to the common supply channel is arrayed; a common collecting channel configured to collect liquid from the plurality of pressure chambers; and a

collection port array in which a plurality of collection ports configured to collect liquid from the common collecting channel is arrayed, wherein the discharge port array and the collection port array extend from a first end of the liquid discharge head to a second end of the liquid discharge head, wherein the plurality of discharge ports in the discharge port array includes a first-end discharge port at the first end and a second-end discharge port at the second end, wherein the plurality of collection ports in the collection port array includes a first-end collection port at the first end and a second-end collection port at the second end, wherein, as seen from a position facing openings of the plurality of discharge ports, an end of an opening of the first-end collection port adjacent to the first end is at a position nearer to the first end than an end of an opening of the first-end discharge port adjacent to the first end, and wherein an end of an opening of the first-end collection port adjacent to the second end is at a position nearer to the first end than the end of the opening of the first-end discharge port adjacent to the first end.
16. The liquid discharge head according to claim 15, wherein the plurality of supply ports in the supply port array includes a first-end supply port closest to the first end and a second-end supply port closest to the second end, and wherein at least one of: an end of an opening of the first-end supply port adjacent to the second end is at a position nearer to the first end than the end of the opening of the first-end discharge port adjacent to the first end, and an end of an opening of the second-end supply port adjacent to the first end is at a position nearer to the second end than the end of the opening of the second-end discharge port adjacent to the second end.
17. The liquid discharge head according to claim 15, wherein an end of an opening of the second-end collection port adjacent to the second end is at a position nearer to the second end than an end of an opening of the second-end discharge port adjacent to the second end.
18. The liquid discharge head according to claim 17, wherein an end of an opening of the second-end collection port adjacent to the first end is at a position nearer to the second end than the end of the opening of the second-end discharge port adjacent to the second end.